

Bocconi

COMPUTER SCIENCE

Code 30424 a.y. 2024-2025

Lesson 11

Creating macros in Excel



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Objectives of the lesson

- Understand the concept and the use of macros
- Learn how to use Excel's macro recorder
- Learn how to create custom commands to launch a macro

What is a macro?

- A macro in Excel is a series of commands and instructions that you can group together as a single command to accomplish a task automatically
- It can be created by writing a set of instructions (lines of code) in VBA (Visual Basic for Applications) or by using the Excel built-in macro recorder
- When using the macro recorder you have to carry out the precise sequence of operations that the macro have to perform when executed
- Purpose of a macro:
 - Automate repetitive tasks
 - Save time by performing complex operations quickly
 - Reduce human error in spreadsheet work



To Remember

- **Enable macros:**
in Excel macros are disabled by default for security. Enable them via:
File > Options > Trust Center > Trust Center Settings > Macro Settings
- **Save the file as Macro-Enabled Workbook:**
to save an Excel file containing a macro you have to use .xlsm format
- **Use VBA for complex tasks:**
while recording is good for simple tasks, VBA is needed for more complex tasks. It is also possible to edit and/or extend with more instructions the script created by the macro recorder

Our case

- The **Sales** worksheet contains data of sales over **4 years** in **184 stores** in different states of the US
- Our objective is to make so that is possible to show a **compact view** of the table containing just the “**Top 15**” or “**Bottom 15**” stores based on the ranking of the average value of sales
- Before creating the macros we have to complete the dataset with all the required calculations

Our case: main tasks

- Complete columns from **F** to **J** and rows **187** and **188** with all the required calculations
- Create 2 Macros to filter “Top 15” and “Bottom 15” stores (hiding columns not needed and applying different colors to the visible headings)
- Create a Macro to restore the original table (both, the content and the layout)
- Create 3 buttons to execute each macro more easily

Development

- Verify the **Macro Settings** in Excel (recommended option: “Disable all macros with notification”)
- Open the file “**30424 lesson 11 - case 1.xlsx**” and use “**save as**” command to change the extension to **.xlsm**
- Complete the dataset with all the required calculations
- Access the Excel macro recorder via the command **Record Macro**.
Note: to avoid runtime errors all operations in the following slides must be executed following exactly the listed order

Macro #1: Top_15

Start recording the 1st macro saving it as Top_15

1. Select the **Worksheet** tab "Sales"
2. Apply the Filter tool and on the **Ranking** column **filter** numbers that are **Less than 16** (Number Filters > Less than...) then **sort** ascending
3. Write in cell **A1** "Top 15 stores"
4. Apply **Green fill** to headers **A1:E1**
5. Select **Col. B** and **F:J** and **Hide** them (then select cell A1)

Stop recording the macro Top_15

Macro #2: Restore

Start recording the 2nd macro saving it as **Restore**

6. Select the **Worksheet** tab "**Sales**"
7. Select **Col. A:K** and **Unhide** them
8. Write in cell **A1** "**Stores**"
9. Apply **Dark Blue** fill to headers **A1:E1**
10. Remove **Filter** tool
11. Sort by **Col. A**

Stop recording the Macro **Restore**

Macro #3: Bottom_15

Start recording the 3rd macro saving it as Bottom_15

12. Select the **Worksheet** tab "Sales"
13. Apply the Filter tool and on the **Ranking** column **filter** numbers that are **Top 15** (Number Filters > Top 10... > set 15 items) then **sort** ascending
14. Write in cell **A1** "Bottom 15 stores"
15. Apply **Red fill** to headers **A1:E1**
16. Select **Col. B** and **F:J** and **Hide** them (then select cell A1)

Stop recording the macro Bottom_15

Running macros

As an alternative to the command “Macros > View Macros > Run” we can create three buttons to launch our macros:

- **Insert** the three **shapes** (green, blue and red)
- **Write** in each the **name** of a macro
- **Assign** to each shape the respective **macro**
- **Set** the shapes **properties** to “Don’t move or size with cells”

Tools to learn with today's lesson

- Macro recorder
- Basic VBA editing
- Assigning macros to shapes



Book references

Excel for students in economics and finance, 2nd edition:

Chapter 7

Excel workbook, 4th edition:

Exercises on the functions and tools of the lesson